MODEL QUESTION PAPER

TED (15)-3094 Reg.No ………………………….

(REVISION-2015) Signature ………………………..

THIRD SEMESTER DIPLOMA EXAMINATION IN POLYMER TECHNOLOGY-

TECHNOLOGY OF ELASTOMERS

[Time: 3 hours

(Maximum marks: 100)

PART-A

(Maximum marks: 10)

Marks

I Answer the following questions in one or two sentences. Each question carries 2 marks.

1. What is sulphur blooming?
2. Write the dosage of sulphur and accelerator in CV cure system.
3. Define cure time.
4. What happens to hardness when plasticizers are added to rubber compound?
5. Which is more tacky? NR or SBR compound. (5x2=10)

PART-B

(Maximum marks: 30)

II Answer ***any five*** of the following questions.Each question carries 6 marks.

1 Explain primary, secondary and delayed action accelerators with examples. (6)

2 Define compounding and explain the objectives of compounding. (6)

3 Illustrate a cure graph of rheometer and mark the parameters maximum torque, minimum torque, scorch time, optimum cure time. (6)

4 Design an NR based compound having a hardness of 60 shore A using HAF as filler. (6)

5 Write suitable non- sulphur curing system for olefin rubbers . (6)

6 Write down the classification of blowing agents with examples. (6)

7 Describe the difference between electrical heating and steam heating. (6)

PART-C

(Maximum Marks: 60)

(Answer ***one full*** question from each unit. Each question carries 15 marks.)

UNIT-I

III a) State the function of accelerators and their classification with examples (8)

b) What are non- black fillers? Write a short note on inorganic fillers used in rubber compounding. (7)

OR

IV a) List the special purpose additives and their functions. (8)

b) Describe the procedure to determine Iodine number of carbon black. What is its significance? (7)

UNIT-II

V a) State the importance of base polymer in a compound design. (6)

b) Explain the type of crosslinks formed and their effect on vulcanisate properties for sulphur and non-sulphur curing systems. (9)

OR

VI a) Explain the mastication, masterbatch, gum compound and filled compound. (8)

b)How can you compare the curing of NR and SBR compound? Write the dosage of curing system based on sulphur . (7)

UNIT-III

VII a) Explain Moving dierheometer(MDR) to determine the cure characteristics of rubber compound. (8)

b) Explain the working of an injection moulding machine with a diagram. (7)

OR

VIII a) Distinguish the cure graph for NR and SBR. (8)

b) What is over cure? What are the different observations we get from a cure graph during over cure? (7)

UNIT-IV

IX a) Describe the elements to be considered for preparing a formulation and developing in to a product (8)

b) Design an NR compound formulation with silica filler to get 60 shore A hardness. (7)

OR

X a) Write a formulation for heat resistant pressure cooker gasket. Justify the ingredients. (8)

b) Describe the factors to be considered for compounding a product having good ozone resistance. (7)
